A brief introduction to:

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Statistics in Space

Building the intuition for spatial and spatiotemporal generalized linear mixed models (GLMMs)

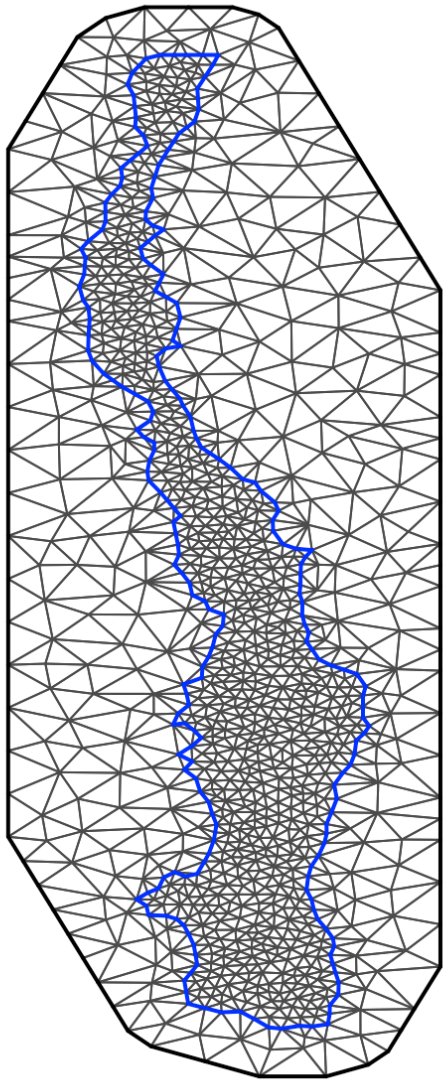
Generalized linear models and maximum likelihood

- Simulated quantile-based residuals and evaluating residual spatial autocorrelation

GLMMs and the Laplace approximation

Spatial and spatiotemporal GLMMs

- Gaussian Fields, Matérn covariance
- The SPDE method



A triangulated Chesapeake Bay

Follow along in R: `R/statistics_in_space_intro.r`

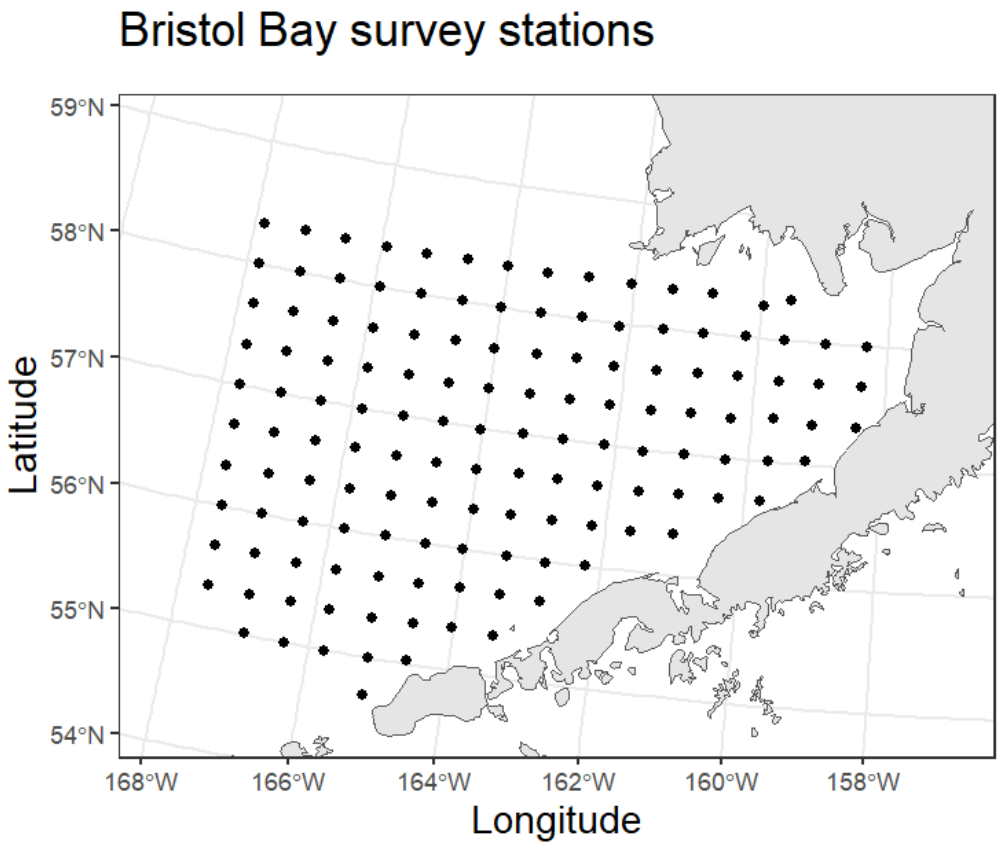
Generalized linear models

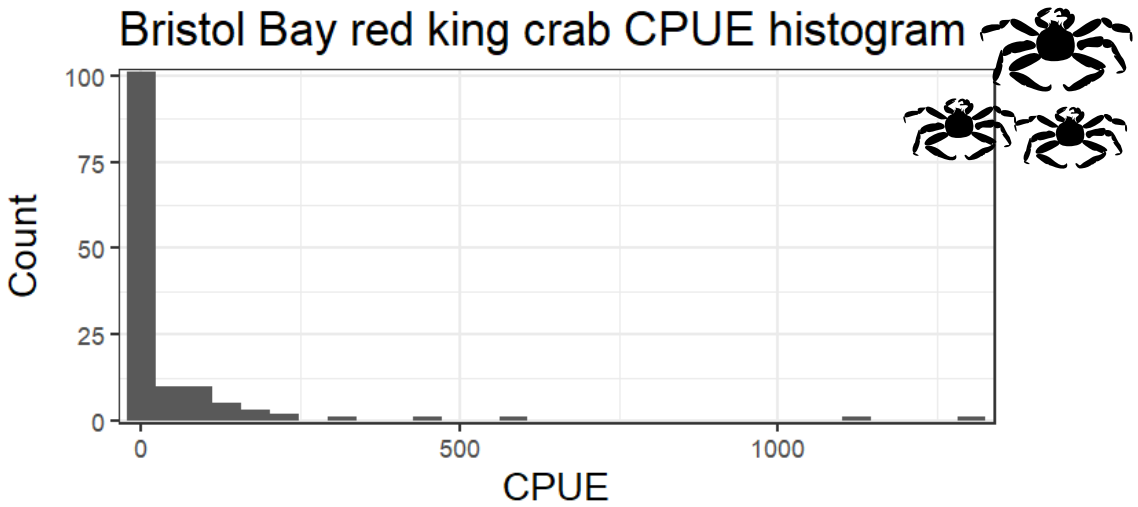
GLMs allow for modeling observations drawn from diverse *probability distributions*

- Gaussian, Poisson, negative binomial, Tweedie etc

Question: What was the mean catch per unit effort (CPUE) of red king crab in Bristol Bay in the 2010 NMFS bottom trawl survey?

- CPUE must be ≥ 0 and commonly right-skewed
- Tweedie probability distribution could be good *data model* for CPUE
  - Useful for continuous positive data with zeros present (e.g., biomass/numbers density)





Generalized linear models

Approach: Fit an intercept-only, Tweedie GLM to the data

The model requires specifying:

1. A **data model**

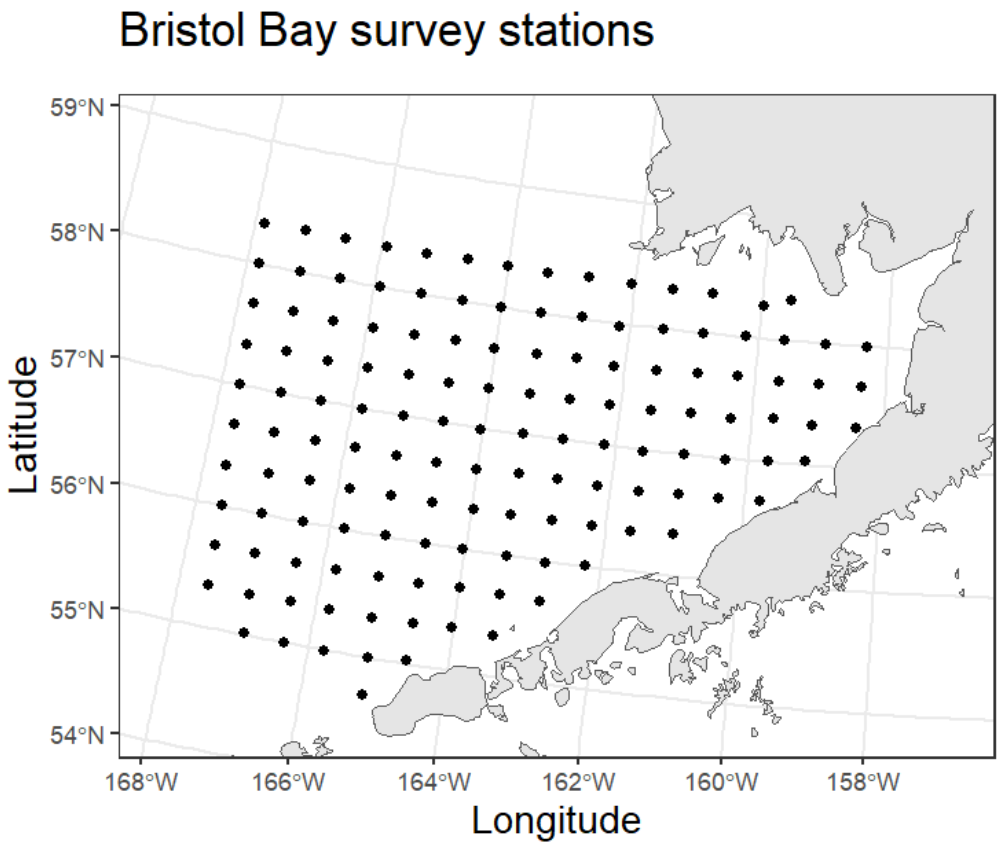
$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

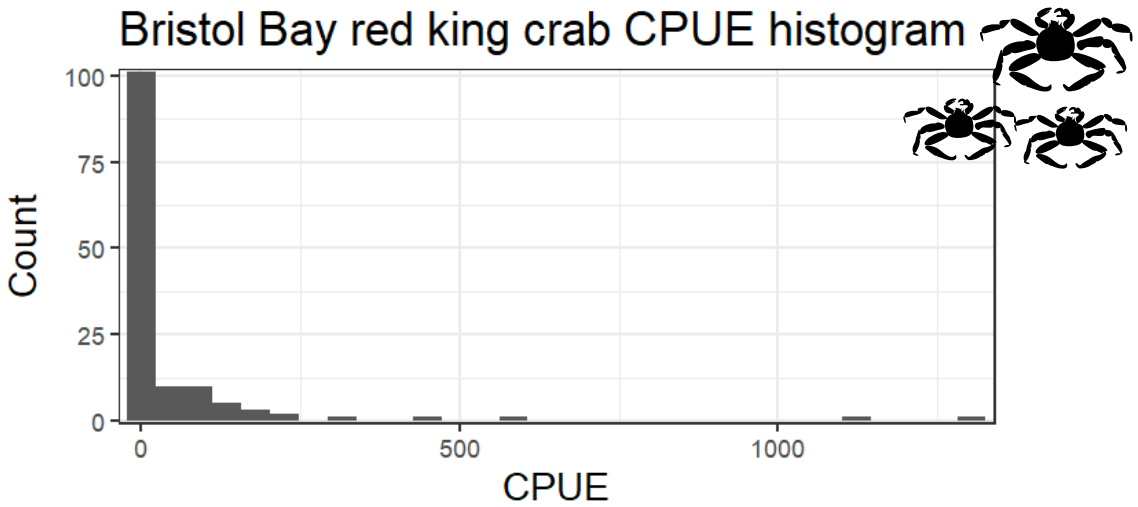
2. A **linear predictor** with link function

$$\log(\mu_s) = \beta_0$$

β_0, ϕ, p are parameters that need to be estimated

- β_0 is log-CPUE at sampling location *s*
- ϕ and p are parameters characterizing Tweedie shape



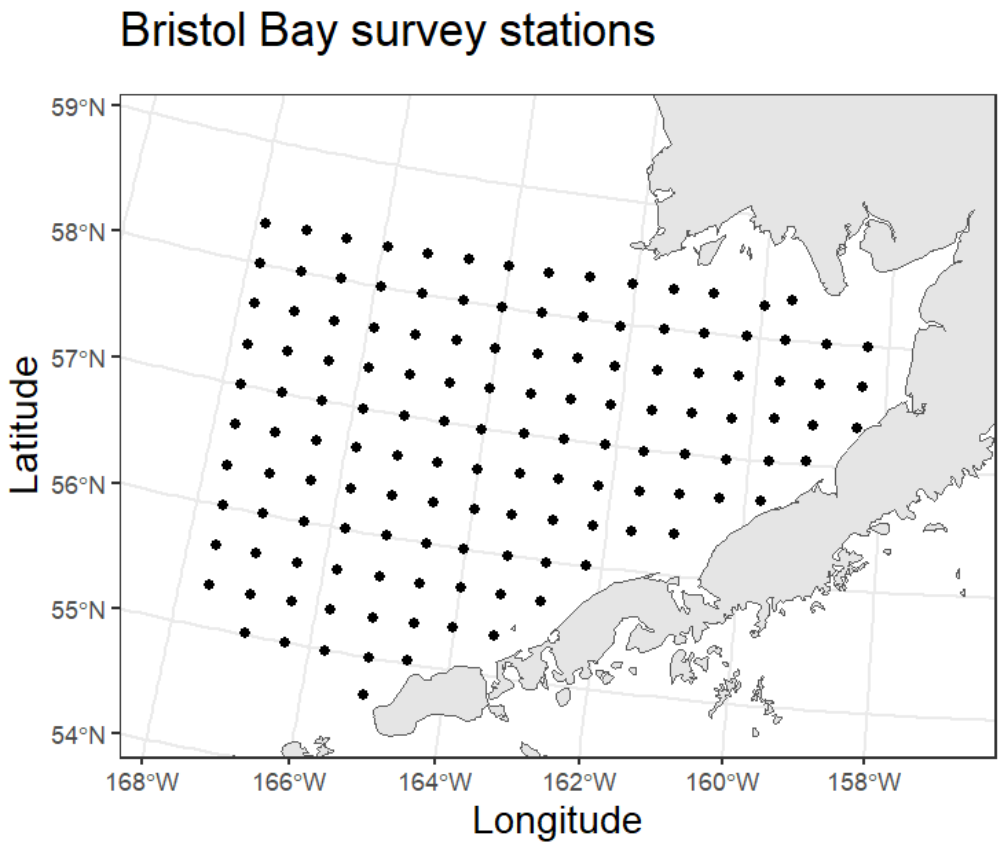


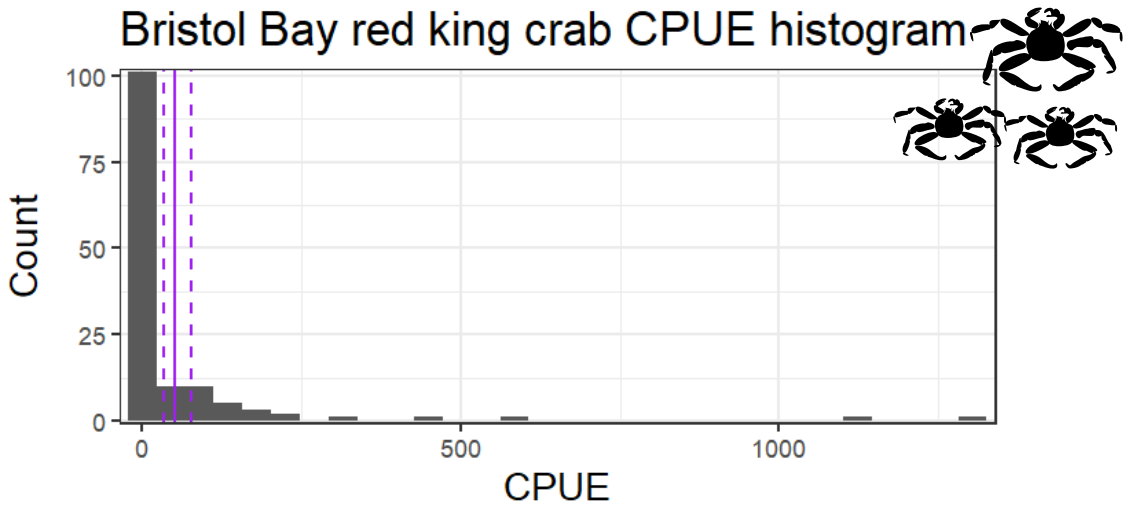
Generalized linear models

Fitting the model in *sdmTMB* via maximum likelihood estimation (MLE)

```
> # intercept only GLM-----------------------------------
> mod1 <- sdmTMB(cpue ~ 1,
+                family = sdmTMB::tweedie(),
+                spatial = "off",
+                data = rkc_cpue_2010_df)
> summary(mod1)
Model fit by ML ['sdmTMB']
Formula: cpue ~ 1
Mesh: NULL (isotropic covariance)
Data: rkc_cpue_2010_df
Family: tweedie(link = 'log')

Conditional model:
            coef.est coef.se
(Intercept)     3.91    0.21

Dispersion parameter: 39.57
Tweedie p: 1.52
ML criterion at convergence: 328.084

See ?tidy.sdmTMB to extract these values as a data frame.
```

The answer to our question is $\mu_s = exp(\beta_0) = 50.12$ crabs per sq. km





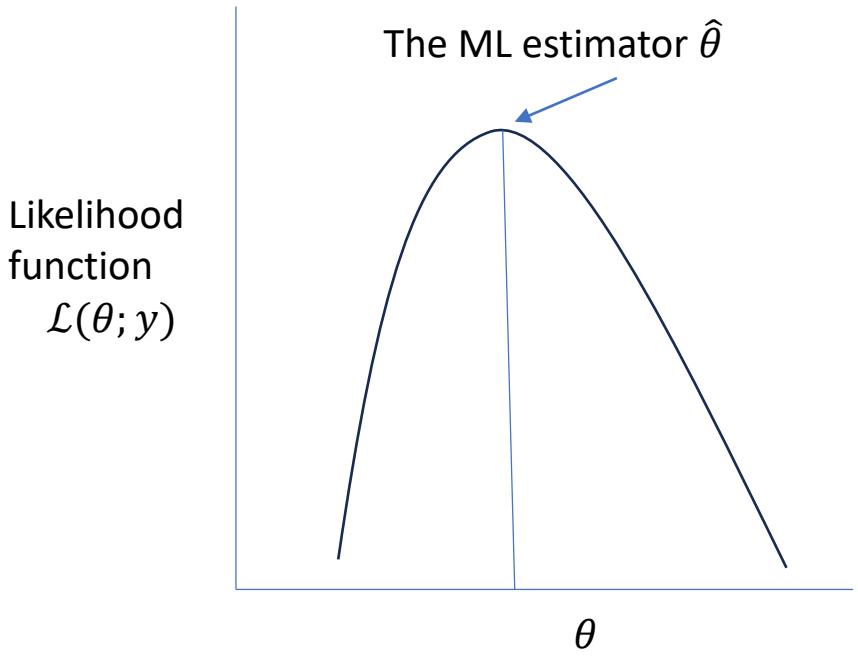
Spare a moment for maximum likelihood estimation

MLE for GLM involves:

1. Specifying a probability model for the data (Tweedie here)

2. Using the model to calculate the likelihood $\mathcal{L}(\theta; y)$ of observing each observation y given a set of parameters θ

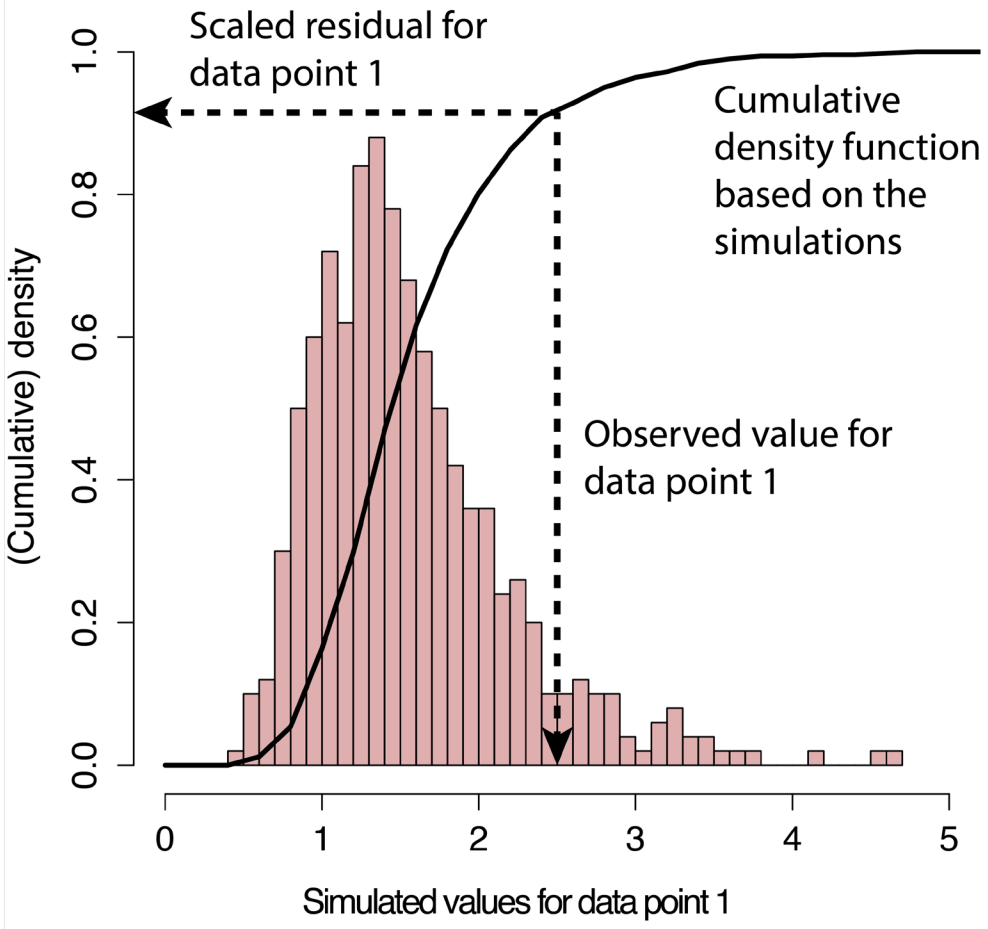
3. Identifying the parameters $\hat{\theta}$ that maximize the joint likelihood of all observations using an optimization algorithm
   - In practice, we minimize the joint negative log-likelihood of the data



MLE assumes residual independence

- **Violations of this assumption can bias inference and uncertainty**

DHARMa residuals



Are residuals independent?

Simulation-based quantile residuals:

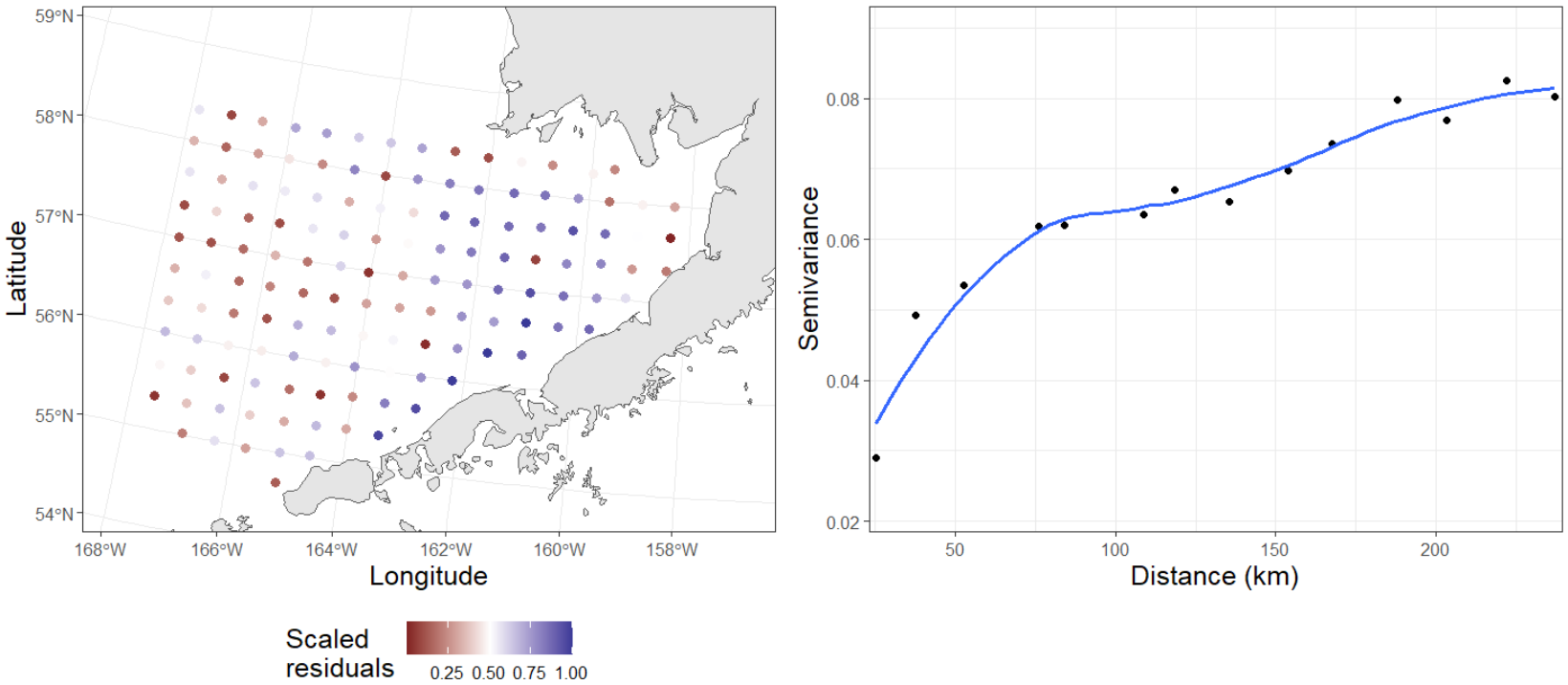
1. We simulate observations from the fitted model to provide the expected distribution for each observation.

2. The fraction of simulated observations $\leq$ the observation is the **quantile residual**

Well-behaved DHARMa residuals will be uniformly distributed and are **scaled residuals**

Works for diverse response families and random effect structures

Back to crabs. Are residuals independent?

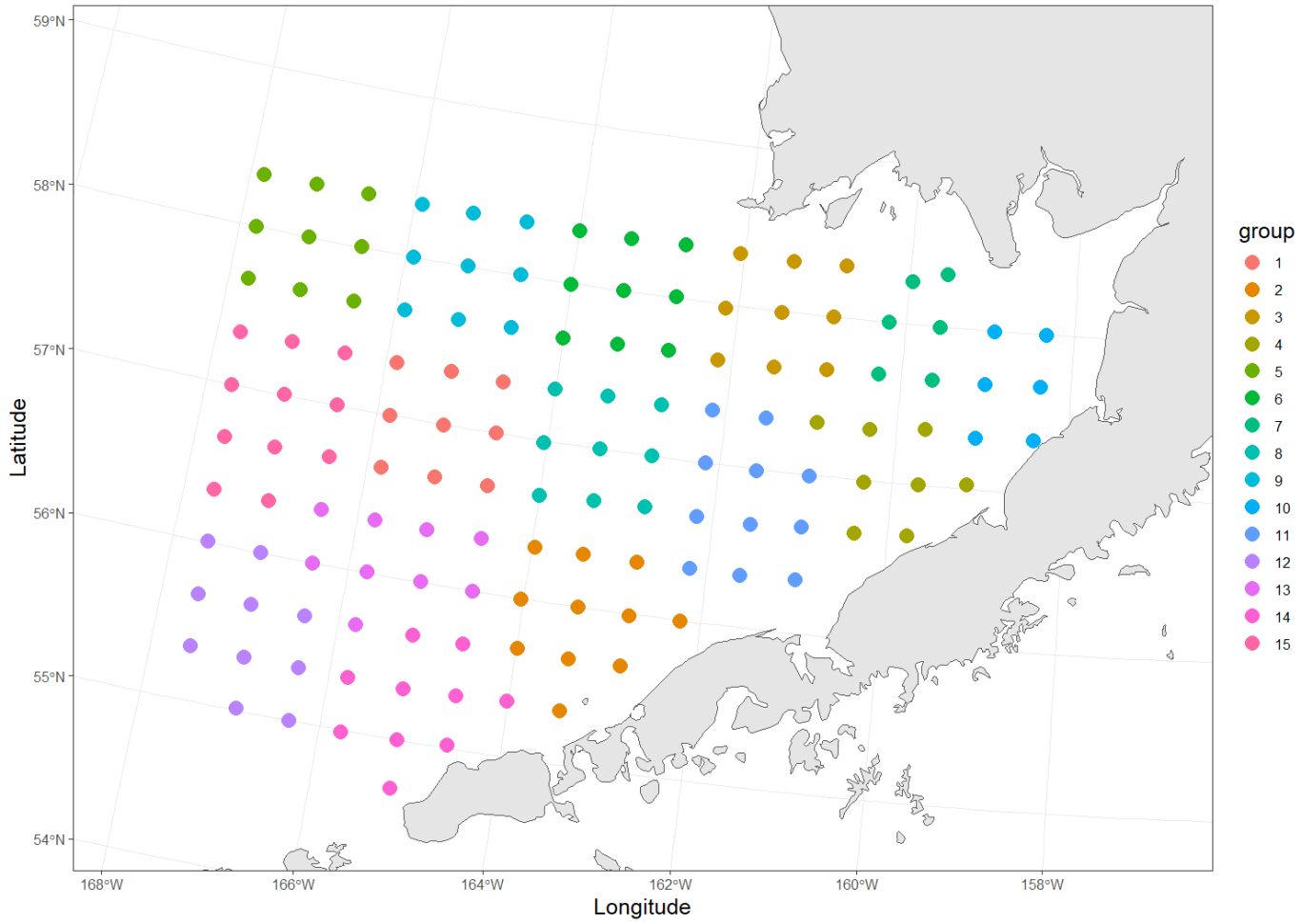


Sample variogram analysis to evaluate residual spatial autocorrelation from Tweedie GLM

- Evaluates variance among observations at varying distances (semivariance), where lower values indicate higher correlation
- More complex models will better explain residual variation and result in semivariance quickly plateauing

Suggests that residual spatial autocorrelation begin to diminish at ~75 km but does not plateau

Introducing the GLMM



A simple option is to add spatial structure for groups of observations

Model group-level deviations from the overall mean using **random intercept** α_g

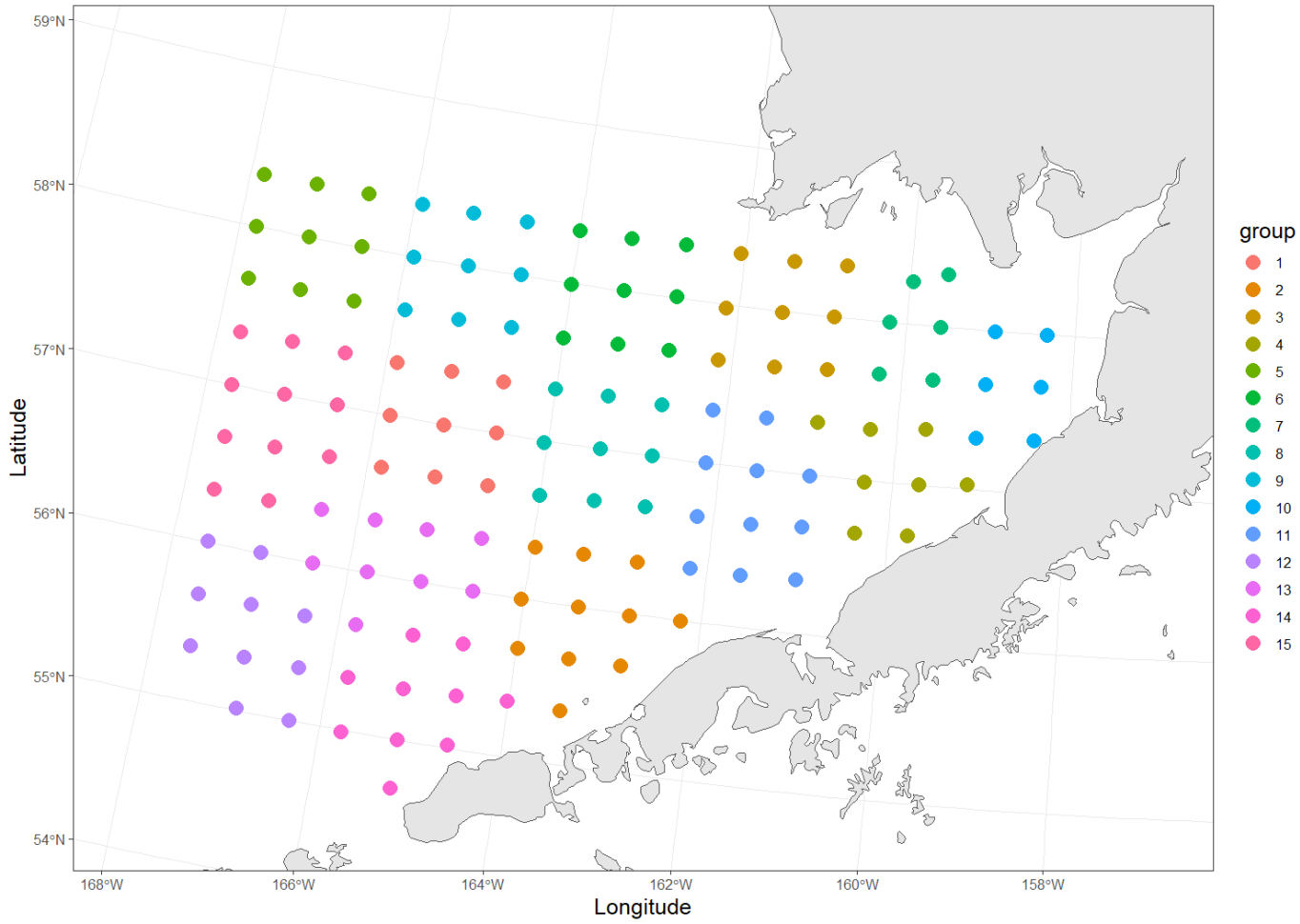
With a random intercept, our model becomes a GLMM

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

$$\log(\mu_s) = \beta_0 + \alpha_g$$

$$\alpha_g \sim N(0, \sigma_g^2)$$

Introducing the GLMM



What is a random intercept?

- A group-level "correction" or "modification" of the population-level intercept β_0
- Random intercepts capture within, but not between, group correlations
- Clunky but useful approach to account for residual spatial autocorrelation

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

$$\log(\mu_s) = \beta_0 + \alpha_g$$

$$\alpha_g \sim N(0, \sigma_g^2)$$

Maximum marginal likelihood estimation for GLMM

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

$$\log(\mu_s) = \beta_0 + \alpha_g$$

$$\alpha_g \sim N(0, \sigma_g^2)$$

Evaluating the likelihood for GLMMs requires computing an integral over random effects

- The general structure of this integral is

$$\mathcal{L}(\theta; Y) = \int P(Y|\theta, \epsilon) P(\epsilon|\theta) d\epsilon$$

θ - a vector of parameters (e.g.,$\beta_0, \phi, p, \sigma_g^2$)

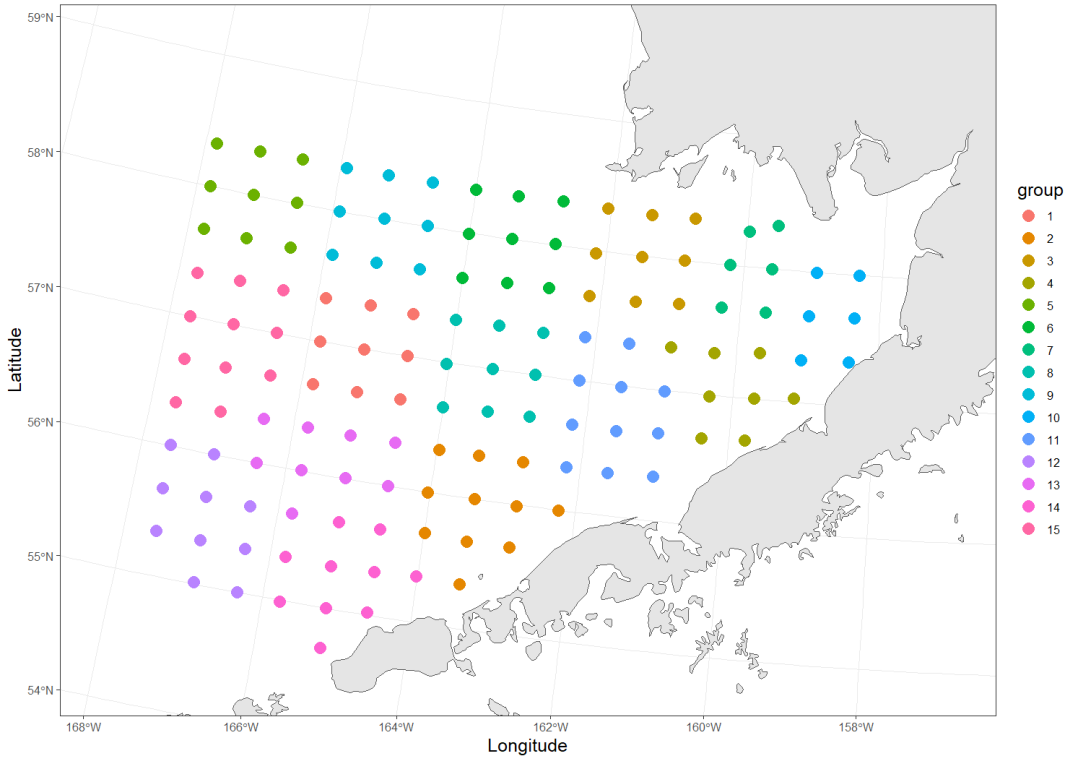
Y – observed data

ϵ – random effects (e.g., α)

Template Model Builder (TMB), the C++ software that *sdmTMB/tinyVAST* use to estimate models, evaluates this integral efficiently via the **Laplace approximation**

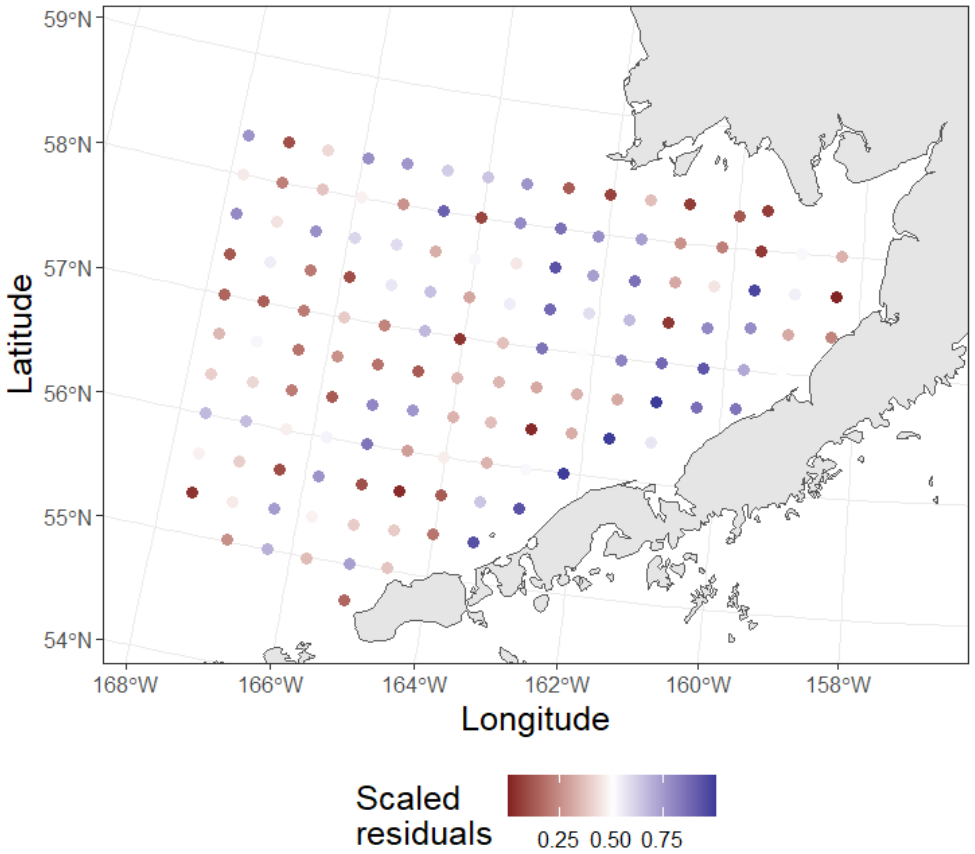
- TMB uses **automatic differentiation** to calculate gradients of the log-likelihood surface
- An optimizer uses these gradients to find the parameters that maximize the marginal log-likelihood $\mathcal{L}(\theta; Y)$

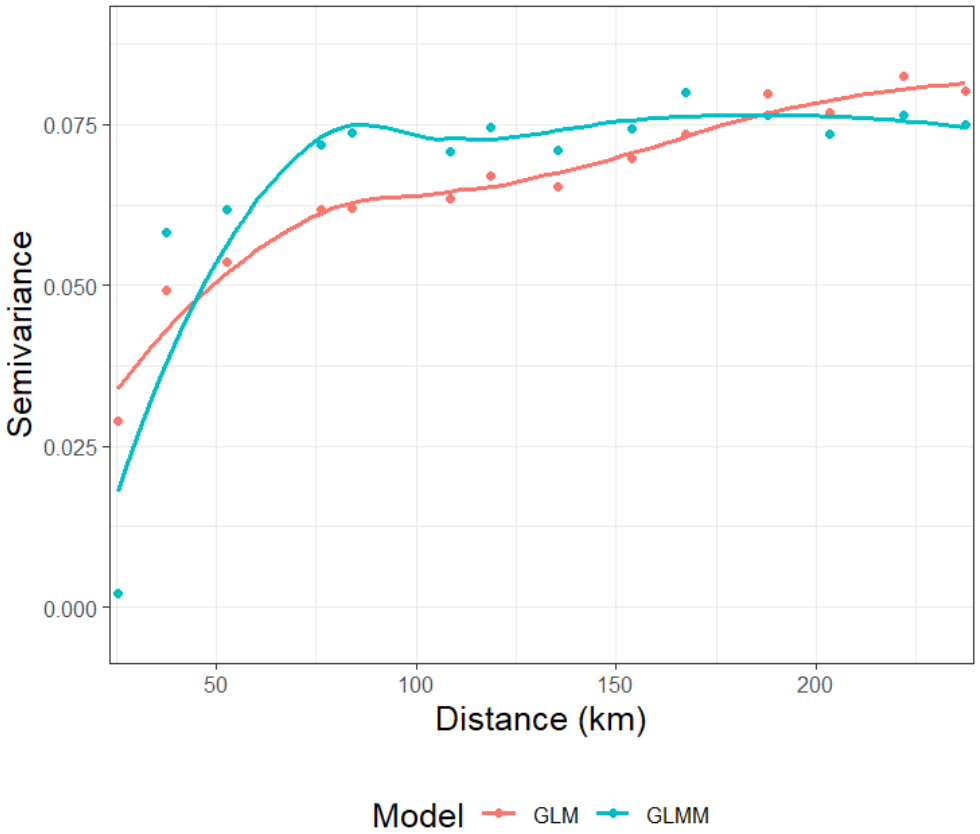
Fitting simple GLMMs in *sdmTMB*




```
> ## fit the GLMM
> mod2 <- sdmTMB(cpue ~ 1 + (1|group),
+                family = sdmTMB::tweedie(),
+                spatial = "off",
+                data = rkc_cpue_2010_df)
> mod2
Model fit by ML ['sdmTMB']
Formula: cpue ~ 1 + (1 | group)
Mesh: NULL (isotropic covariance)
Data: rkc_cpue_2010_df
Family: tweedie(link = 'log')

Random intercepts and/or slopes:

Conditional model:
     Groups        Name    Variance    Std.Dev.
      group (Intercept)        7.87        2.81

Conditional model:
            coef.est coef.se
(Intercept)      1.8    1.04

Dispersion parameter: 31.13
Tweedie p: 1.38
ML criterion at convergence: 306.906

See ?tidy.sdmTMB to extract these values as a data frame.
```

Evaluating GLMM residual spatial autocorrelation





Inclusion of random intercept reduces residual spatial autocorrelation relative to GLM

- More evenly distributed spatial residuals
- Residuals are less well correlated at shorter distances, less obvious spatial structure on map

GLM vs. GLMM predictions

The GLMM with improved model residual diagnostics has a substantially lower mean CPUE estimate:

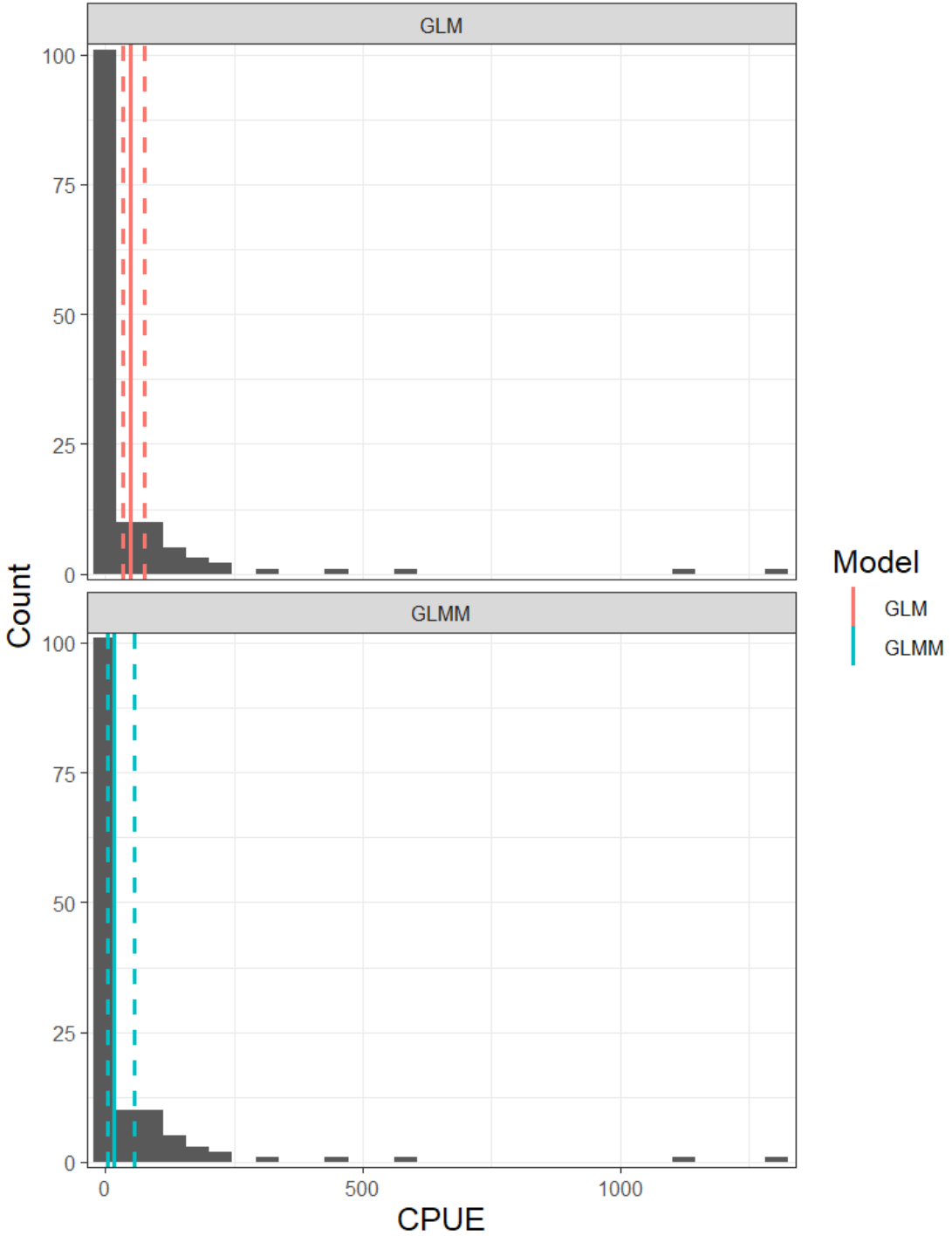
- **GLM** estimated mean = **50.1** (95% CI: 33.2 - 75.8)
- **GLMM** estimated mean = **17.0** (95% CI: 5.1 - 57.2)

The hacky random intercept improved residual diagnostics but the approach is limited:

1. The GLMM model assumes between-group residual independence despite spatially continuous correlations among observations

2. We have no way to interpolate CPUE with uncertainty at unmeasured locations

How can we do better? **Spatial GLMM**



Intuition for Gaussian fields and spatial GLMM

Generically, suppose we collect samples of some variable over a continuous spatial domain

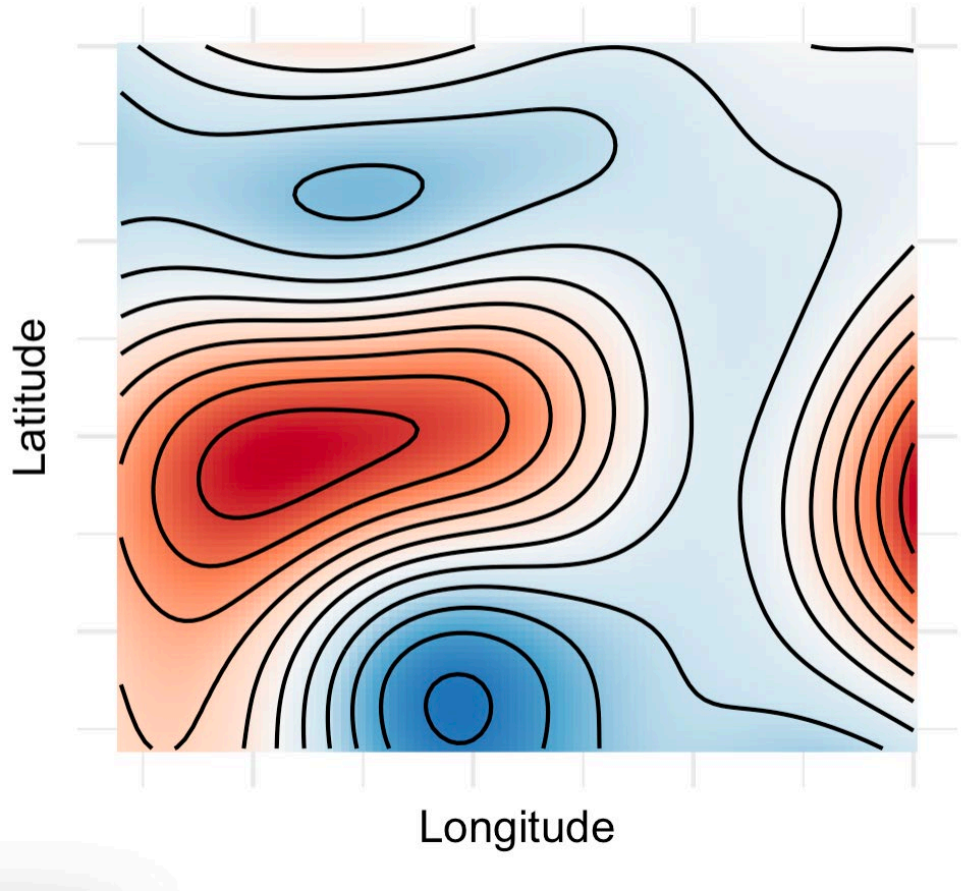
These samples are draws from a underlying *stochastic process*

- If we repeat sampling under the exact same conditions, we will end up with different realizations of the same process

If we assume observations are Gaussian, then we have a Gaussian Field

In our spatial GLMM

- We model unexplained (by fixed effects) variability in CPUE using a residual term, $\boldsymbol{\omega}$
- $\boldsymbol{\omega}$ is modeled as a GF with mean 0 and covariance $\boldsymbol{\Sigma}_{\omega}$



Generic example of a 2D Gaussian Field

$$\boldsymbol{\omega} \sim \mathrm{GF}(\mathbf{0}, \boldsymbol{\Sigma}_{\omega})$$

Spatial GLMMs

Modeling the covariance matrix $\boldsymbol{\Sigma}_\omega$ allows for making predictions with uncertainty at unmeasured locations – *kriging*

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

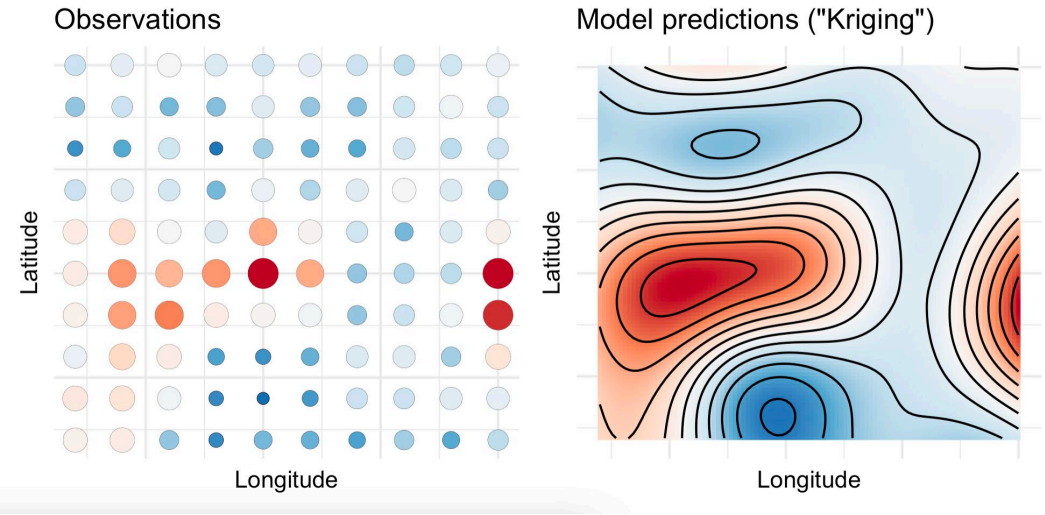
$$\log(\mu_s) = \beta_0 + \omega_s$$

$$\boldsymbol{\omega} \sim \mathrm{GF}(\mathbf{0}, \boldsymbol{\Sigma}_\omega)$$

The elements of covariance matrix $\boldsymbol{\Sigma}_\omega$ are covariances between all $\boldsymbol{\omega}$, too many to estimate

Instead, we calculate covariances based on

1. Euclidean distances between observations
2. A function representing how spatial covariance decays with distance, the **Matérn covariance function**



Generic example of kriging interpolation

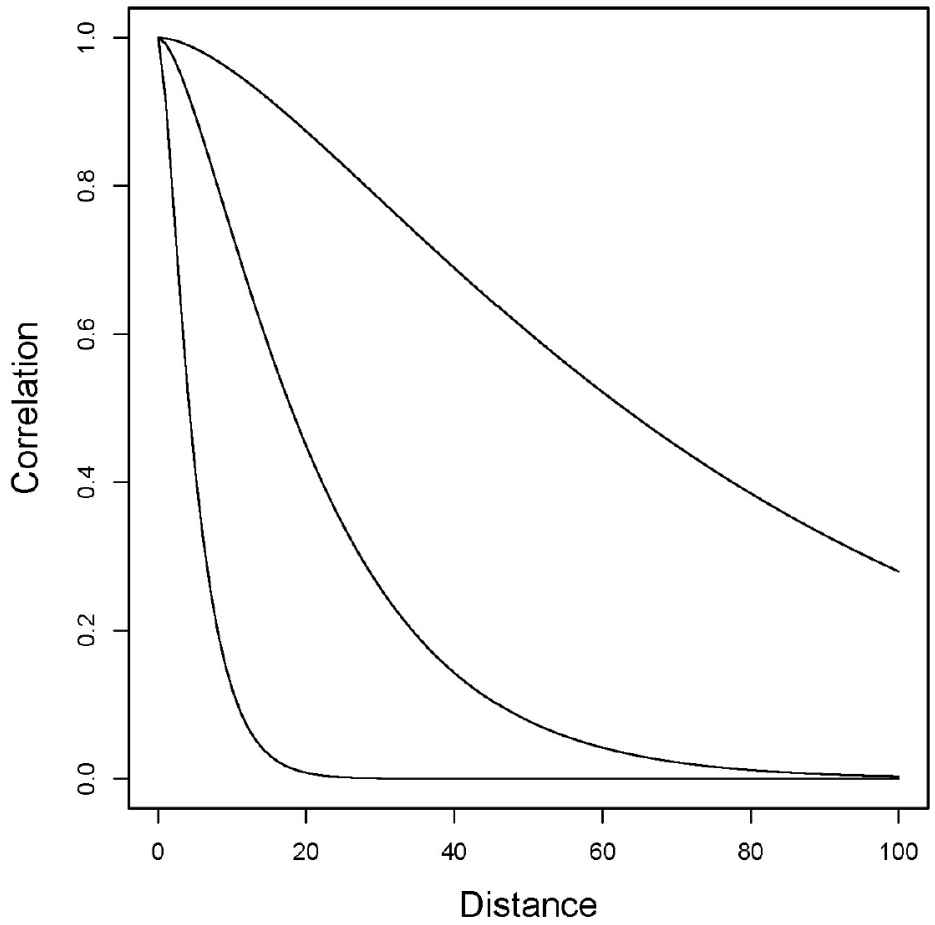
Matérn covariance

In *sdmTMB*, calculating Matérn covariance requires estimating two parameters:

1. κ – a scale parameter where the **spatial range** = $\sqrt{8}/\kappa$
2. σ_m^2 - the marginal variance aka the 'magnitude of random field "wiggles" '

Given these parameters and the Euclidean distance between observations we can fill-in the spatial covariances of $\boldsymbol{\Sigma}_\omega$

An assumption of Matérn covariance is **isotropy** – that correlation decays equally in all directions



Three Matérn correlation functions for different k (Zuur et al. 2017)

Computational shortcuts: GF ⟶ GMRF

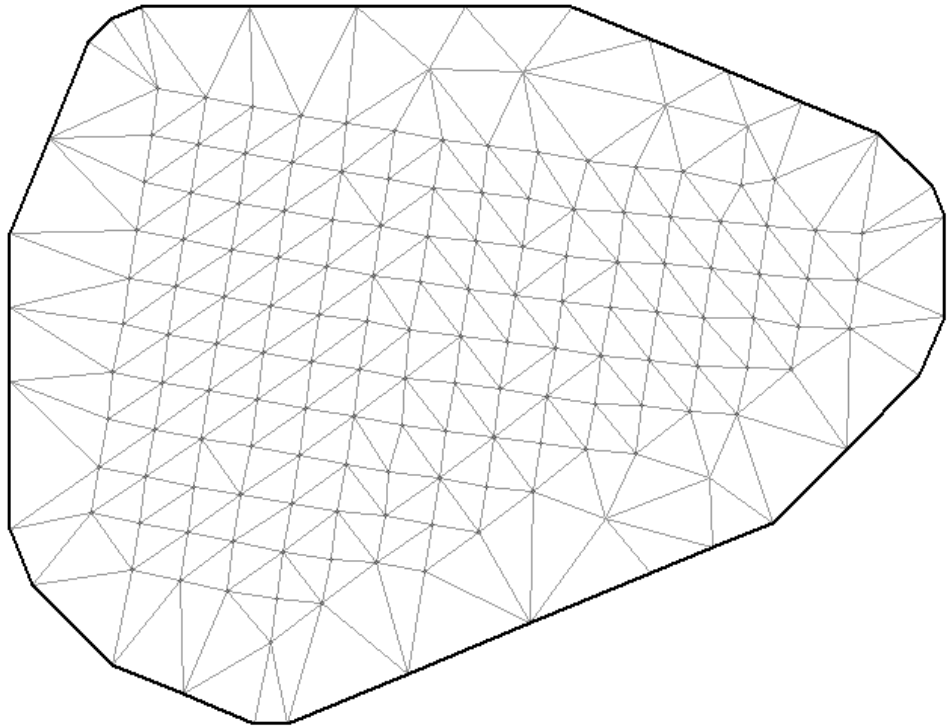
Working with dense covariance matrices like $\boldsymbol{\Sigma}_{\omega}$ is computationally expensive

To improve efficiency, we approximate the GF as a **Gaussian Markov Random Field (GMRF)**

- The precision matrix Q for the GMRF is sparse and easier to work with than the dense covariance

This approximation is done using the **stochastic partial differential equation (SPDE)** approach

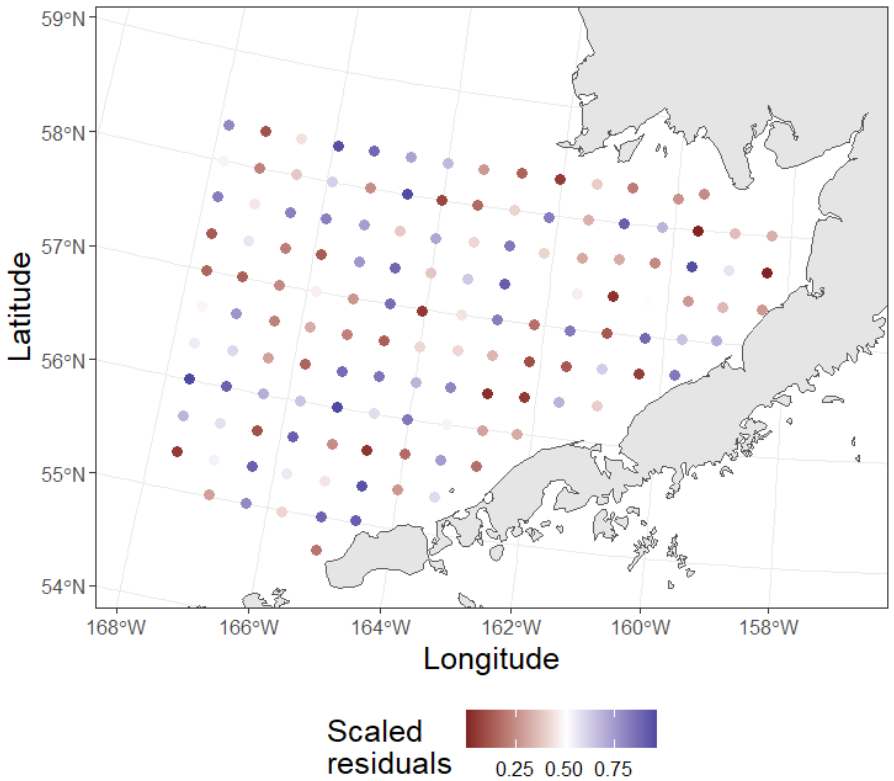
The approximation is performed using a finite element mesh (*"the mesh")* over the spatial domain

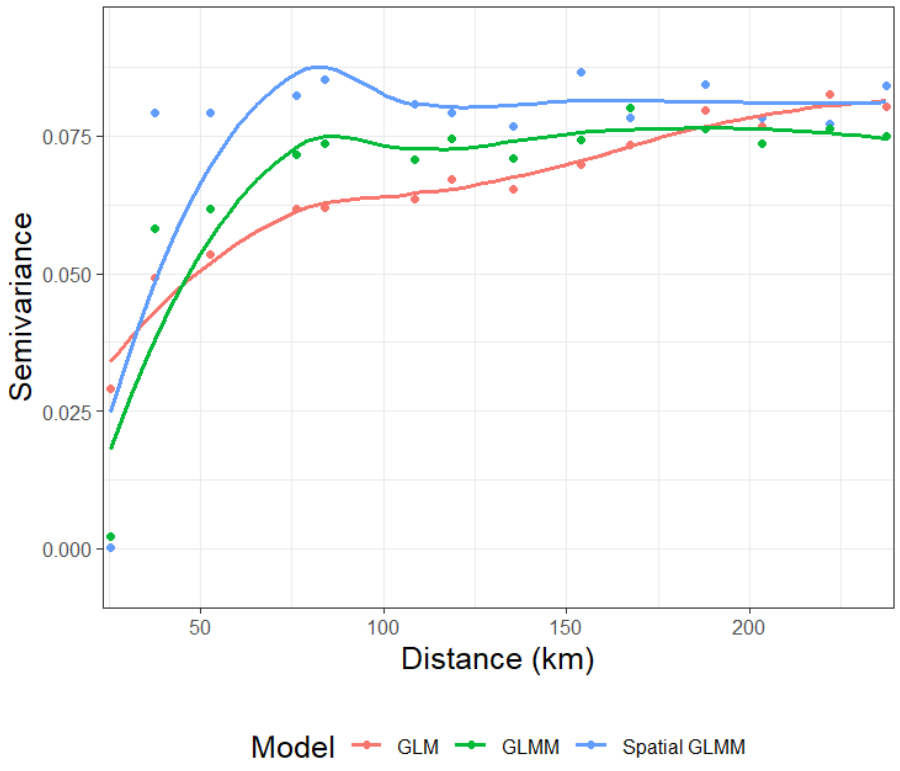


A finite element mesh defined over Bristol Bay

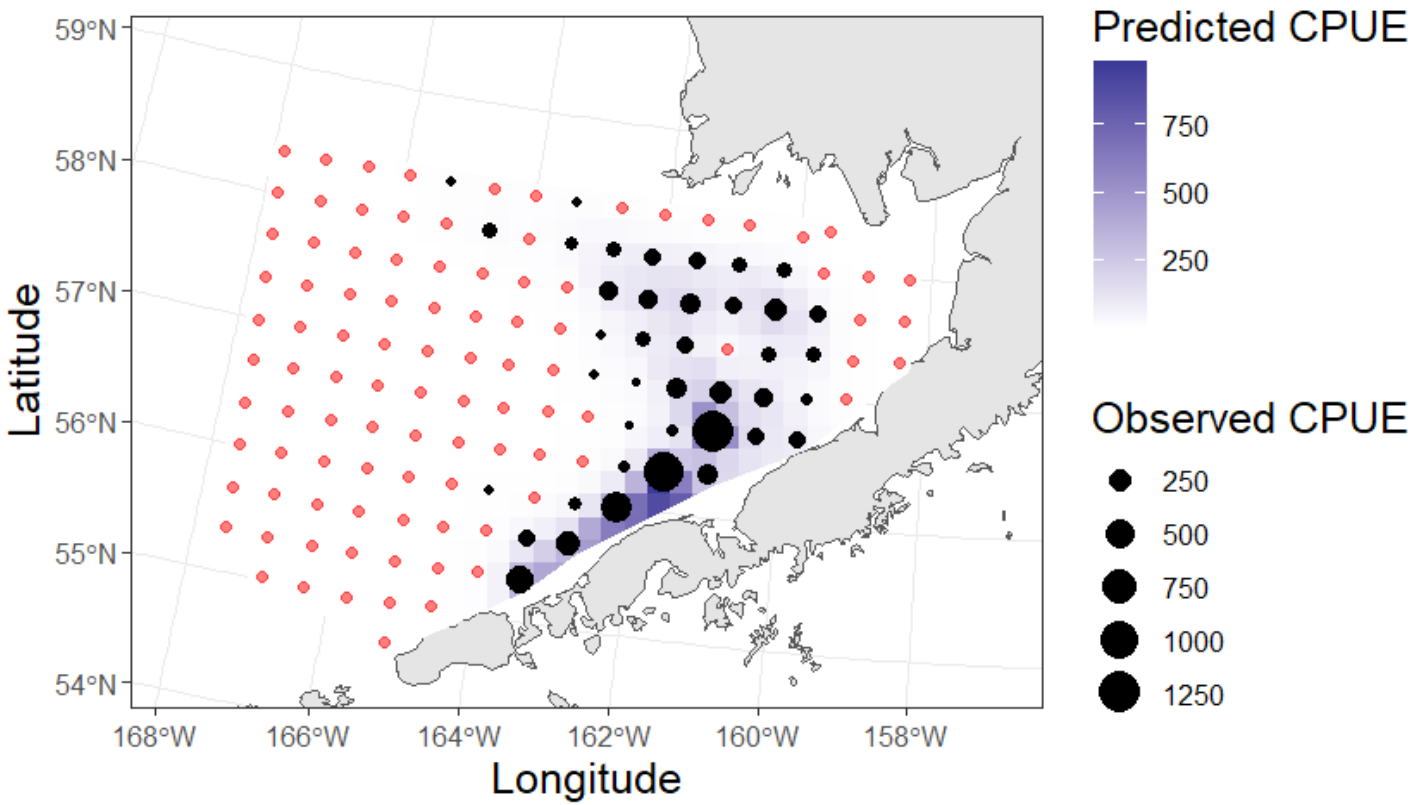
Fitting the spatial GLMM

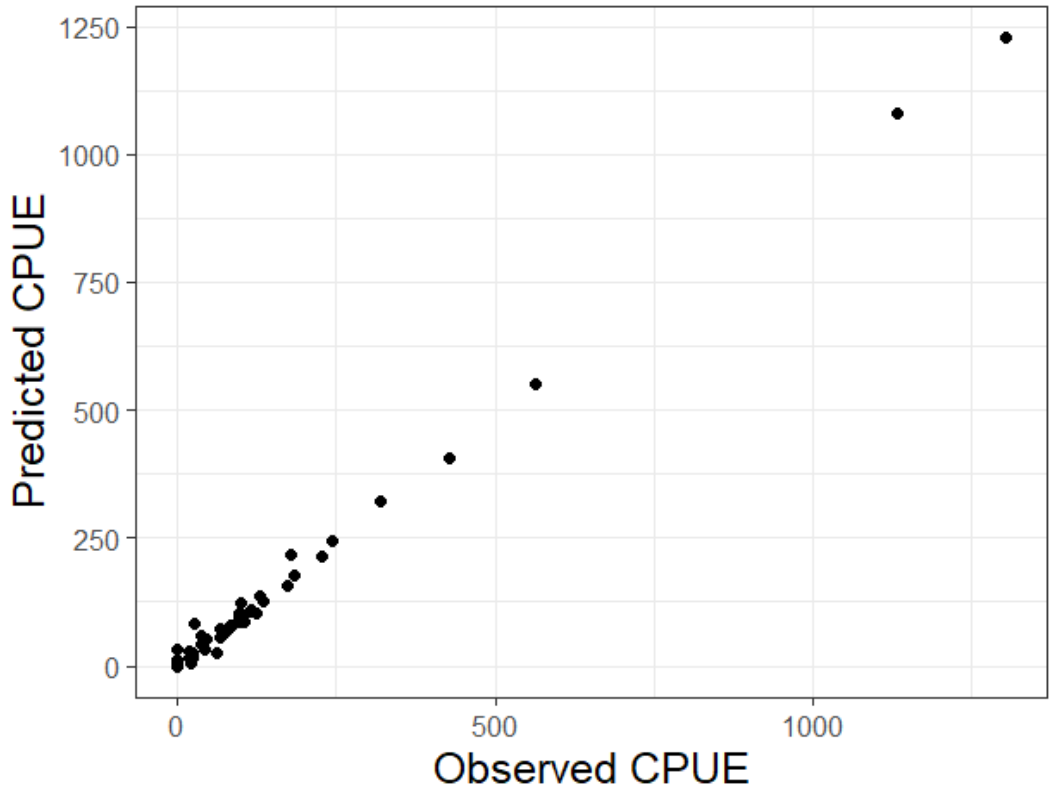
```
> # Spatial GLMM-------------------------------------
> rkc_mesh <- make_mesh(data = rkc_cpue_2010_df,
+                       xy_cols = c("longitude", "latitude"),
+                       type = "cutoff", cutoff = 10)
> mod3 <- sdmTMB(cpue ~ 1,
+                family = sdmTMB::tweedie(),
+                spatial = "on",
+                mesh = rkc_mesh,
+                data = rkc_cpue_2010_df)
```





Fitting the spatial GLMM





Expanding into space and time: IID spatiotemporal random fields

The GMRF estimates latent spatial variability in a residual term

Spatial processes evolve through time

Model spatiotemporal random fields as temporally independent ("iid") or spatiotemporally correlated (AR1/random walk)

- Spatiotemporal correlation structure is required for spatiotemporal interpolation and projection

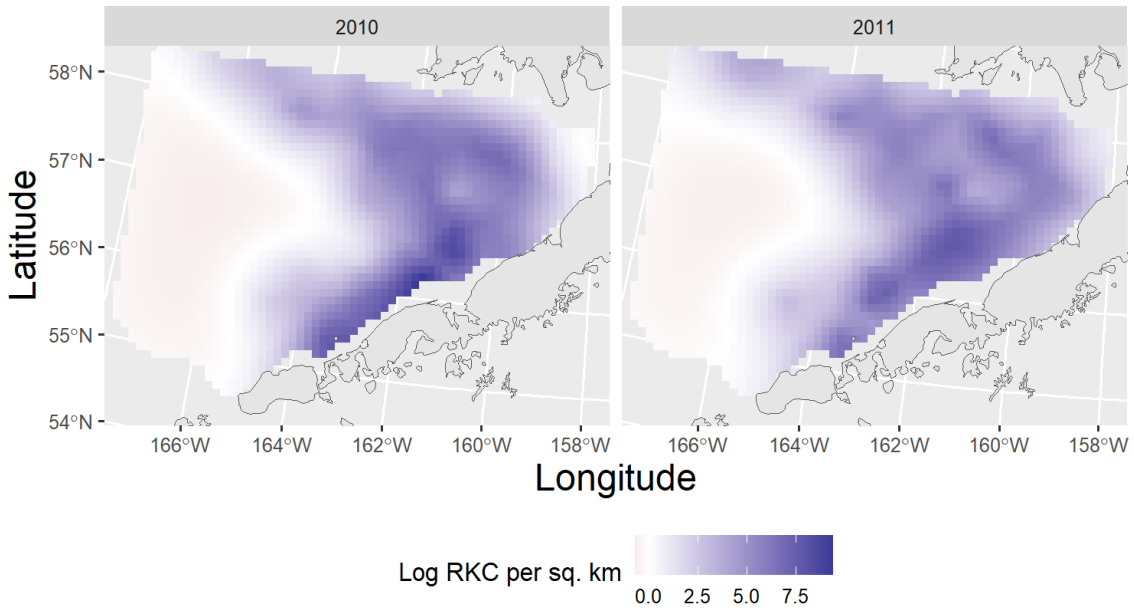
We will explore these correlation structures in detail in the following demonstration and labs

$$\mathrm{CPUE}_{s,t} \sim \mathrm{Tweedie}(\mu_{s,t}, \phi, p)$$

$$\log(\mu_{s,t}) = \beta_0 + \omega_s + \epsilon_{s,t}$$

$$\boldsymbol{\omega} \sim \mathrm{GMRF}(\mathbf{0}, \boldsymbol{\Sigma}_\omega)$$

$$\boldsymbol{\epsilon}_t \sim \mathrm{GMRF}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon)$$



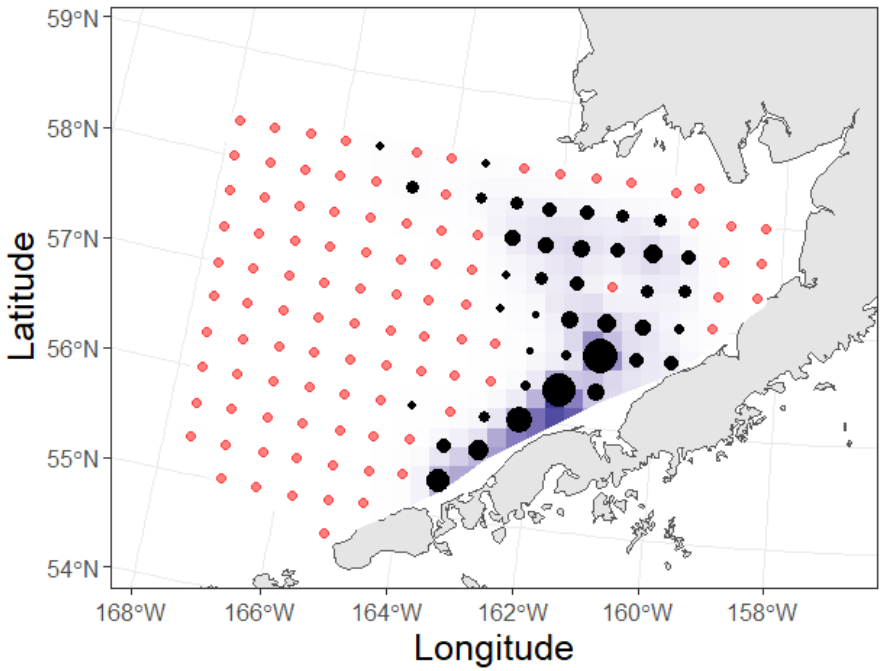
Recap

GLMs are useful for modeling data drawn from diverse probability distributions

- Fitted with MLE that assumes residual independence
- DHARMa residuals/sample variograms

Simple GLMM with random intercepts is useful but limited

- Spatial autocorrelation is continuous, not discrete + no capacity for interpolation/projection
- These and more complex models with random effects/fields fitted using *marginal maximum likelihood estimation*

Spatial and spatiotemporal GLMMs

- Involve estimating Matern covariance; approximating the GF as a GMRF via SPDE method
- Resolves residual spatial autocorrelation to improve inference
- Relies on spatial covariance to skillfully predict